

Data Literacy for the 21st Century: Perspectives from Visualization, Cognitive Science, Artificial Intelligence, and Education

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Abstract

Data literacy is becoming a foundational skillset in our increasingly data-driven society. Fields that rely heavily on data, such as data visualization, cognitive psychology, and artificial intelligence, each contribute unique perspectives in studying data literacy. Yet, these efforts remain siloed. Researchers studying visualization literacy and AI literacy have recently run separate workshops at ACM CHI, each independently calling for interdisciplinary conversations. Such conversations would not only advance the study of data literacy but also inform research agendas within each individual discipline. Building on ACM CHI's track record of attracting diverse researchers across many disciplines, we propose a workshop that connects these literacies through the common ground of data literacy. We invite interdisciplinary perspectives by bringing researchers, practitioners, and students together with expertise from fields including HCI, data visualization, psychology, AI, and learning sciences to develop a holistic framework for understanding, measuring, and teaching data literacy that is grounded in real-world applications.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; Visualization theory, concepts and paradigms.

Keywords

Data literacy, Interdisciplinary workshop, Visualization literacy, Statistical literacy, AI literacy

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1 Motivation

Data drives the modern world. It provides the basis for compelling visualizations that inform decision-making in business and government. Huge datasets are used to train AI models capable of performing an ever-growing variety of complex tasks. How people think with data is a cornerstone of the cognitive science of reasoning and learning. This ubiquity of data makes *data literacy* essential in the 21st century [4, 28]. Data literacy encompasses many competencies, including engaging in the process of inquiry about data (e.g., [34]), thinking critically about data (e.g., [28]) and drawing appropriate inferences from data (e.g., [5]). In essence, data literacy enables a person to *use data to answer real-world questions* [33].

Many disciplines that rely on data contribute unique perspectives on the study of data literacy. As a common tool for communicating data, visualizations offer researchers a window into *how* people extract information and make meaning from data, thereby targeting competencies related to data sharing and communication. Cognitive psychologists study *why* people make errors when interpreting data and what mental models they construct when reasoning about data. AI researchers situate data literacy within the context of machine learning, and focus on how people interpret algorithmic outputs while accounting for bias or other limits in the training data. Learning scientists investigate *how to teach* these complex and interconnected skills effectively. All of these disciplines offer distinctive perspectives on *what it means* to be data literate in an increasingly algorithm-driven world.

In order to construct a holistic framework for understanding, measuring, and teaching data literacy grounded in real-world applications, we are creating this workshop to bridge the related

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literacies (e.g., visualization literacy [2], statistical literacy [16], AI literacy [23]) that rely on data as a foundation. Through interdisciplinary perspectives, we strive to advance both the study of data literacy and the research agendas within each individual discipline.

1.1 Note About Past Workshops

This workshop builds on the success of independently-organized workshops on visualization literacy at CHI 2024 [14] and AI literacy at CHI 2023 [24]. These past workshops demonstrate strong interests in both literacies within and beyond the CHI community. A key outcome of the workshop on visualization literacy emphasized the complex and interdisciplinary nature of the construct [10]. Similarly, one of the future research directions identified in the AI literacy workshop was exploring how to define and situate AI literacy for different audiences and contexts, necessitating interdisciplinary expertise to build effective learning interventions. Recognizing data literacy as common ground between these parallel efforts, we propose this workshop to bridge these research communities and establish a sustained interdisciplinary dialogue.

1.2 Workshop Goals and Topics for Discussion

The goals of our two-session workshop include facilitating interdisciplinary conversations, forming new research agendas, and fostering collaborations. We invite researchers, practitioners, and students with expertise in HCI, visualization, psychology, AI, learning sciences, and other relevant disciplines to contribute to topics including, but not limited to:

- designing formal and informal learning environments to improve data literacy from different perspectives
- developing assessments to measure learners' data literacy in different contexts
- designing tools that uncover the cognitive processes important to be data literate
- formulating a shared vocabulary across disciplines for a joint effort for studying data literacy
- bridging data literacy and individual disciplines to situate data literacy in specific contexts
- forming research agendas within different disciplines through this shared lens of data literacy

2 Length of the Workshop

As per CHI 2026 guidelines, this workshop will be in-person only. We aim to cultivate an interactive and collaborative workshop driven by deep discussions and hands-on activities, drawing on the format of previous workshops a subset of us organized [14, 24]. There will be two sessions, each lasting 90 minutes. The first session will be composed of panel discussions based on the accepted submissions. The topics that emerge during the first session will serve as seed questions for subgroup activities in the second session. We expect 20 to 30 registered participants (more details in Section 7).

3 Organizers

Our organizing committee are equipped with the expertise spanning a wide range of disciplines, such as data visualization (e.g., Ge [6, 7, 10–14], Fan [3, 31], Kay [6, 7, 10–12, 14, 18]), cognitive psychology

(e.g., Fan [1, 9, 30]), AI literacy (e.g., Long [8, 22–24]), and learning sciences (e.g., Horn [15, 19, 20, 29]).

Lily Ge is a Ph.D. candidate in Computer Science and a member of the Midwest Uncertainty Collective at Northwestern University. She studies how people read, create, and critically reason about visualizations (i.e., visualization literacy). She has previously led the investigation of how to assess people's susceptibility to misleading visualizations [11] and ability to encode data for effective visual communication [12]. She has co-organized successful workshops at CHI 2024 [14] and VIS 2024 [27] on the topic of visualization literacy and education.

Michael Horn is a Professor jointly appointed in Computer Science and the Learning Sciences at Northwestern University. His research explores the use of emerging interactive technology in the design of novel learning experiences. His projects include the design and evaluation of a tangible computer programming language for use in science museums and early elementary school classrooms; and the design of multi-touch tabletop exhibits for use in natural history museums. His most recent work has involved the design of a learning platform to support digital music production and computational literacy for middle school and high school students.

Duri Long is an Assistant Professor in the Department of Communication Studies at Northwestern University. Her research investigates how people interact with AI, how to design learning experiences to foster broader AI literacy, and how to model co-creative practices to inform the development of AI. She is the author of the most widely cited paper on AI literacy [23] and co-organized the first CHI workshop on AI literacy [24]. Her work has resulted in numerous public-facing AI literacy resources, including low-cost 'unplugged' activities for AI education [8] and interactive exhibits involving co-creative AI that have been exhibited publicly at museums across the US [22].

Judith Fan is an Assistant Professor of Psychology and, by courtesy, Education and Computer Science, at Stanford University. She directs the Cognitive Tools Lab, which aims to reverse engineer the human cognitive toolkit, especially how people use physical representations of thought to learn, communicate, and solve problems. Her recent work focuses on developing measures of data and visualization literacy, as well as understanding the factors that predict student success and learning outcomes in college-level data science courses.

Matthew Kay is an Associate Professor jointly appointed in Computer Science and Communications Studies at Northwestern University. He works in uncertainty visualization, visualization literacy, and the design of human-centered tools for data analysis. He has organized successful CHI workshops and SIGs on various topics, including visualization grammars [26], transparent statistics [21, 32], designing for uncertainty [17], health informatics [25], and visualization literacy [14]. He co-directs the Midwest Uncertainty Collective (<https://mucollective.northwestern.edu/>).

4 Pre-Workshop Plans

We detail our plans before the workshop, such as publicizing our workshop website and soliciting submissions from potential participants, our review process for selecting participants, and our plans to publish the workshop proceedings.

4.1 Publicize Workshop and Solicit Submissions

If the workshop proposal is accepted to CHI 2026, we plan to immediately release and publicize the call for participation. We will launch a website to distribute updated workshop information, including the call for participation, workshop structure, goals, and schedule (when it becomes available). We will publicize the call for participation through multiple channels. Besides our workshop website (<https://data-literacy-workshop.github.io/CHI26/>), we will distribute it across social media platforms and communities of previous workshops on visualization literacy and AI literacy. Our organizing team has extensive connections across a variety of fields, and can therefore distribute the call to an interdisciplinary community of scholars with diverse backgrounds. Submissions will be due by February 10, 2026, and we will send notification of results by early March.

4.2 Review Process

Because we are prioritizing group discussions and activities in our two-session workshop, we seek diverse voices across different disciplines within our panels and subgroups. We plan to use a juried selection approach to review the paper submissions and select a subset of submissions that workshop organizers can then use to help group participants into panels. Selection by a jury can better ensure the representation of diverse perspectives in reaching our goal of facilitating interdisciplinary conversations on data literacy. The selection jury will consist of a subset of workshop organizers and invited experts. We aim to accept a range of 10 to 20 submissions, which will inform the number and the themes of the panels. Submissions are not required to be anonymous.

4.3 Plans to Publish Workshop Proceedings

We will publish the accepted papers on arXiv and link the papers on the workshop website before the conference. The outputs and artifacts generated as a result of the workshop will also be hosted on the workshop website, which will help facilitate discussions before, during, and after the workshop.

5 Accessibility

All paper submissions should comply with CHI 2026 accessibility guidelines, as emphasized in the call for participation (Section 9). We will also ensure that all workshop related materials used during the workshop and posted on the website after the workshop are accessible. We will do our best to provide accessibility accommodations if requested by participants (e.g., transcription). Outputs generated from collaborative technologies (e.g., Google Docs, Miro) should ensure accessibility by the general public as well as adequate support for asynchronous engagement.

6 Offline Materials for Asynchronous Engagement

For interested individuals who cannot participate synchronously during the workshop sessions, we will host materials from before, during, and after the workshop on the workshop website to support asynchronous engagement. These materials include accepted

papers, notes taken during the panel discussions and group activities, and outputs from the group activities (e.g., affinity mapping, written artifacts). We will also summarize the workshop activities and publicly host a report on the website to further facilitate discussions.

7 Workshop Activities

The workshop will be driven by panel discussions and group activities, rather than traditional paper presentations. The exact topics for panel discussions will depend on the paper submissions we receive and the themes that arise. On the day of the workshop, we would appreciate a room with AV equipment and an adjustable configuration of tables and chairs to facilitate panel discussions and group activities. We propose a workshop schedule for a long workshop (i.e., two 90-minute sessions) below and detail the structure of the two sessions of the workshop.

7.1 Session 1: Panel Discussions (90 minutes)

A tentative set of panel topics may include how to teach data literacy, how to assess data literacy in different contexts, and other related topics on measuring and improving data literacy (see Section 1.2). The exact topics of the panel discussions depend largely on the submissions we receive. We expect to form at most three panels in this session. Each panel will consist of three to five panelists to ensure more productive discussions. The author(s) from each paper will serve as the panelists. One of the organizers will moderate the panels and manage time during and between panels. We will provide post-it notes for participants to take notes during the panel discussions. The noted topics of interest will be revisited during the group activities. We will also use collaborative technologies such as Google Docs and Miro to support note-taking throughout the workshop, which can be readily accessible online after the workshop.

7.2 Session 2: Group Activities (90 minutes)

The first 30 minutes will be an affinity mapping activity. We will first assemble attendees into smaller groups, where they will use Miro to list ideas that emerged from the panel discussions or add new ideas that they would like to discuss further. Each subgroup will receive a guiding question to tackle, such as challenges related to assessment, intervention, or other topics of interest from the first session. We will then form a large group with all attendees to synthesize themes from each smaller group into a shared document. The emergent themes will be saved and revisited during the subgroup activities.

The next 40 minutes will be subgroup activities. The participants will be invited to choose a topic of interest based on the emergent themes from the affinity mapping activity. Possible group activities include working groups to form concrete research plans or a strategic planning group to brainstorm ways to continue interdisciplinary conversation surrounding data literacy (e.g., organizing informal meetups in between conferences). Participants will start with seed questions such as: What are some research questions related to this topic of interest? What are some challenges in addressing the related problems? How should a research team start to

tackle such challenges? Each group will work together in a shared document to draft concrete artifacts as they work on their activity.

The next 15 minutes will be sharing with the large group. We will reconvene, and each subgroup will share with the large group for about 5 minutes. During the process of sharing, attendees will be encouraged to take notes through collaborative technologies and preserve artifacts that result from these activities.

The last 5 minutes will be reflecting on insights from the workshop as a whole. We will close by summarizing what we accomplished during the workshop and encouraging participants to continue the conversations beyond the workshop.

8 Post-Workshop Plans

We plan to build on the conversations that started during the workshop to collaboratively publish a workshop report, either as a conference paper or a summary report, similar to what we did previously for the visualization literacy [10] and AI literacy¹ workshops. We will collect interest from participants to continue the conversation via online methods (e.g., Zoom) and co-author this output paper. We will also form an online community with the participants for furthering conversations and facilitating potential collaborations. The format of this online community will depend on the preferences of participants, but could take the form of a Slack group or a mailing list. We will post workshop materials on the website soon after the workshop. These materials would include notes from panel discussions, outputs from group activities, a reflection on the workshop, and any established channels for post-workshop communication. We will disseminate related workshop materials to relevant communities (e.g., HCI, visualization, AI, learning sciences, psychology, education) as an effort to broaden participation and engagement.

9 Call for Participation

This in-person, two-session workshop aims to facilitate interdisciplinary conversations around understanding, measuring, and teaching data literacy. We welcome diverse perspectives from fields including but not limited to data visualization, cognitive psychology, AI, and learning sciences. Some questions of interest include:

- How can we quantify data literacy?
- How can we take an interdisciplinary approach in teaching data literacy?
- How can we design and develop interventions to reach a broader audience?
- How can we bridge the related literacies (e.g., visualization literacy, statistical literacy, AI literacy) to enrich the research agendas within each individual discipline?

The workshop will largely be driven by participant panels based on submissions received. More details are on our workshop website: <https://data-literacy-workshop.github.io/CHI26/>. A jury of experts will review the submissions to select a subset that can contribute diverse perspectives on the topic of data literacy. Submissions should be 2-8 pages (excluding references), in single-column ACM Master Article Submission Template, and comply with CHI 2026 paper accessibility guidelines. To make paper submissions, please fill out

this form <https://forms.gle/ZW7Zko1t4GEqob5o8> by February 10, 2026 (AoE). Decision notification can be expected around early March. Accepted papers will be published on arXiv and on our workshop website. We expect 20 to 30 registered participants.

At least one author of each accepted submission must attend the in-person workshop and all participants must register for the workshop. Outputs from the workshop, including panel discussions and group activities will be made available on our workshop website, facilitating further discussions within and beyond the CHI community.

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¹Click to see the post-workshop report from the AI literacy workshop.

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